

Servlet Training Plan (3 Hours per Day)

Total Duration: **18 Hours**

Goal: **Understand servlet fundamentals required for Spring MVC**

Day 1 – Web Architecture & Servlet Fundamentals

Session 1 (Concept – 1 Hour)

Topics

- Web Application Architecture
- Role of a Web Server vs Application Server
- Servlet Container concept
- How **Apache Tomcat** works
- Introduction to **Jakarta Servlet API**

Concept Flow

Browser



Tomcat Server



Servlet Container



Servlet



Response

Session 2 (Setup – 1 Hour)

Topics

- Install **Apache Tomcat**
- Configure Tomcat in **Eclipse IDE**
- Create Maven Web Project
- Project Structure

src/main/java

src/main/webapp

WEB-INF

Session 3 (Hands-on – 1 Hour)

Practice

Create first servlet

```
@WebServlet("/hello")
public class HelloServlet extends HttpServlet {
    protected void doGet(HttpServletRequest req, HttpServletResponse resp) {
        resp.getWriter().println("Hello Servlet");
    }
}
```

Run application in browser.

Day 2 – Servlet Lifecycle & HTTP Methods

Session 1 (Concept – 1 Hour)

Topics

- Servlet lifecycle

init()

service()

destroy()

- What happens when server starts
 - Difference between GenericServlet and HttpServlet
-

Session 2 (Coding – 1 Hour)

Topics

- Implement lifecycle methods
- Logging lifecycle events

Example

```
public void init() {
    System.out.println("Servlet Initialized");
}
```

Session 3 (Hands-on – 1 Hour)

Topics

- HTTP Methods

doGet()

doPost()

Mini Exercise

Create servlet that handles:

GET → Display message

POST → Save data

Day 3 – Form Handling

Session 1 (Concept – 1 Hour)

Topics

- HTML forms
- GET vs POST
- Request parameters

`request.getParameter()`

Session 2 (Coding – 1 Hour)

Build **Student Registration Form**

HTML form

Name

Email

Course

Servlet receives data.

Session 3 (Hands-on – 1 Hour)

Create:

StudentServlet

Tasks

- Read form data
 - Print response
 - Validate fields
-

Day 4 – Request Dispatcher & JSP Integration

Session 1 (Concept – 1 Hour)

Topics

RequestDispatcher

forward()

include()

Difference:

Forward vs Redirect

Session 2 (Coding – 1 Hour)

Example

```
request.setAttribute("name", name);
```

```
request.getRequestDispatcher("success.jsp").forward(request,response);
```

Session 3 (Hands-on – 1 Hour)

Create mini flow

Form → Servlet → JSP

Display student details on JSP.

Day 5 – Session Management

Session 1 (Concept – 1 Hour)

Topics

Session tracking techniques

Cookies

Hidden Fields

URL Rewriting

HttpSession

Focus on:

HttpSession

Session 2 (Coding – 1 Hour)

Example

```
HttpSession session = request.getSession();
```

```
session.setAttribute("username", name);
```

Session 3 (Hands-on – 1 Hour)

Mini project

Login → Dashboard

Session stores logged-in user.

Day 6 – Mini Project (Servlet CRUD)

Session 1 (Design – 1 Hour)

Design small application

Student Management System

Features

Add Student

View Students

Delete Student

Architecture

JSP → Servlet → DAO → Database

Session 2 (Coding – 1 Hour)

Create servlets

AddStudentServlet

ListStudentServlet

DeleteStudentServlet

Session 3 (Hands-on – 1 Hour)

Integrate with database using JDBC.

Run full application.

Servlet Projects

These projects gradually introduce **HTTP handling, sessions, MVC structure, and database integration** while running on a servlet container like **Apache Tomcat** using the **Jakarta Servlet API**.

The difficulty increases step-by-step.

1. Hello Servlet Application

Goal: Understand servlet setup and HTTP request handling.

Features

- Create a servlet
- Handle doGet()
- Print response to browser

Concepts

- Servlet mapping
 - Request → Response lifecycle
-

2. Student Registration Form

Goal: Learn form handling.

Features

- HTML form
- Submit student details
- Display submitted data

Concepts

- request.getParameter()
 - Form validation
 - POST method
-

3. Simple Calculator Web App

Goal: Handle input and processing logic.

Features

- Two numbers input

- Select operation (+ − × ÷)
- Display result

Concepts

- Servlet processing logic
 - Conditional statements
 - Form handling
-

4. Login & Logout System

Goal: Understand session management.

Features

- Login page
- Session creation
- Logout functionality

Concepts

- HttpSession
 - Session attributes
 - Session invalidation
-

5. Student Management System (CRUD with JDBC)

Goal: First database-integrated servlet project.

Features

- Add student
- View students
- Update student
- Delete student

Concepts

- JDBC integration
 - DAO pattern
 - MVC architecture
-

6. Employee Management System

Goal: Strengthen CRUD architecture.

Features

- Employee registration
- Employee list
- Edit employee
- Delete employee

Concepts

- MVC structure

Example structure

controller

dao

model

util

7. Online Feedback System

Goal: Handle form submission and data storage.

Features

- Submit feedback
- Store feedback in database
- Display all feedback

Concepts

- Request forwarding
 - JSP display
-

8. Shopping Cart (Session Based)

Goal: Implement session-based state management.

Features

- Product list
- Add to cart
- Remove from cart
- View cart

Concepts

- Session storage
 - Collections in session
 - Cart management
-

9. File Upload Application

Goal: Handle file input from browser.

Features

- Upload profile image
- Store file on server
- Display uploaded file

Concepts

- Multipart request
 - File handling
-

10. Mini Blog System

Goal: Build a small MVC web application.

Features

- Create post
- View posts
- Delete posts
- Comment system

Concepts

- Full MVC structure
 - Database interaction
 - Multiple servlets
-

11. User Registration & Login System

Goal: Build a basic authentication flow.

Features

- User registration
- Login validation

- Password storage in DB
- Session-based login

Concepts

- Sessions
 - JDBC authentication
 - Form validation
-

12. Online Quiz Application

Goal: Build a dynamic question-answer system.

Features

- Display questions
- Submit answers
- Calculate score
- Show results

Concepts

- Session tracking
 - Dynamic forms
 - Result calculation
-

13. Product Inventory Management

Goal: CRUD operations for inventory.

Features

- Add product
- Update stock
- Delete product
- Product list

Concepts

- DAO pattern
 - MVC architecture
 - JDBC CRUD operations
-

14. Contact Management System

Goal: Manage contacts like a small CRM.

Features

- Add contact
- Search contact
- Edit contact
- Delete contact

Concepts

- Search queries
 - MVC architecture
 - Request forwarding
-

15. URL Shortener (Basic Version)

Goal: Understand URL mapping logic.

Features

- Enter long URL
- Generate short URL
- Redirect to original link

Concepts

- Redirects
 - Database mapping
 - Unique code generation
-

16. Task Management System (To-Do App)

Goal: Build a productivity tool.

Features

- Create task
- Mark task complete
- Delete task
- Task list

Concepts

- MVC structure
 - JDBC
 - Status handling
-

17. Online Bookstore Catalog

Goal: Create product browsing functionality.

Features

- List books
- View book details
- Search books
- Add to cart

Concepts

- Servlet routing
 - Session cart
 - JSP integration
-

18. Attendance Management System

Goal: Simulate a training or school attendance tracker.

Features

- Add students
- Mark attendance
- View attendance report

Concepts

- Date handling
 - Database relationships
 - Reporting
-

19. Online Poll/Voting System

Goal: Handle user voting.

Features

- Create poll

- Vote for option
- Display results

Concepts

- Request parameters
 - Prevent duplicate voting
 - Data aggregation
-

20. Blog Comment System

Goal: Extend the blog idea from project 10.

Features

- View blog posts
- Add comments
- Delete comments
- Show comments per post

Concepts

- One-to-many relationships
 - Servlet routing
 - MVC structure
-

Final GitHub Repository Structure (20 Projects)

Recommended repository name

java-servlet-projects

Structure

java-servlet-projects

```
|
├── 01-hello-servlet
├── 02-student-registration
├── 03-calculator-app
├── 04-login-session
├── 05-student-crud
├── 06-employee-management
├── 07-feedback-system
├── 08-shopping-cart
└── 09-file-upload
```

- └— 10-mini-blog
 - |
 - └— 11-user-auth-system
 - └— 12-online-quiz
 - └— 13-product-inventory
 - └— 14-contact-manager
 - └— 15-url-shortener
 - └— 16-task-manager
 - └— 17-bookstore-catalog
 - └— 18-attendance-system
 - └— 19-online-poll
 - └— 20-blog-comments
-

What These 20 Projects Prepare You For

After completing them you will understand:

- Servlet architecture
- MVC design pattern
- Session management
- JDBC integration
- Web application flow